SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 –Agile Sprint Planning & User Story Workshop

Course Code:	CSA1017	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – Agile Sprint Planning & User Story Workshop
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
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Date:	20/10/2026	**Duration:**	60 minutes

Code of Conduct

I Guru charan M/192524434 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Guru charan M

1. Project Vision and Objectives

1.1 Project Vision

To provide manufacturers with a secure, reliable, and scalable platform that brings equipment health, maintenance history, preventive schedules, failure-risk indicators, and operational performance into one place—helping teams act before failures occur, reduce unplanned downtime, control maintenance costs, and improve equipment utilization.

1.2 Problem and Opportunity

Reactive repairs and fixed maintenance schedules can cause avoidable downtime, unnecessary servicing, delayed production, and higher operating costs. A centralized platform can combine equipment readings, asset records, maintenance events, and configurable thresholds to help maintenance teams prioritize work using evidence rather than relying only on calendar-based servicing.

1.3 Objectives and Success Measures

Objective	Proposed measure / indicator	Why it matters
Monitor equipment performance	Percentage of in-scope assets with current readings; data freshness	Makes asset condition visible.
Track maintenance activities	Percentage of work orders with status, owner, timestamps, and history	Improves traceability and accountability.
Identify potential failures	Alert precision/recall after validation against labelled events; lead time to failure	Measures usefulness of risk indicators.
Schedule preventive maintenance	Percentage of due tasks scheduled before their due date	Improves planning discipline.
Notify maintenance teams	Notification delivery rate and acknowledgement time	Supports timely response.
Generate analytics	Report completeness, refresh time, and dashboard availability	Supports informed decisions.
Improve utilization and reduce downtime	Baseline vs. post-deployment unplanned downtime and asset utilization	Connects platform use to business outcomes.

Note: Numeric targets should be agreed after collecting a baseline. Predictive performance cannot be responsibly claimed until the platform has representative equipment data and validation results.

1.4 Scope

In scope	Out of scope for the first sprint
Asset register and plant/line hierarchy; equipment health readings; maintenance history; preventive schedules; risk alerts; notifications; dashboards and reports; role-based access; multi-plant support.	Production-grade predictive model deployment, integration with every machine/PLC/SCADA vendor, full ERP integration, mobile applications, and enterprise-wide rollout. These may be delivered in later releases.

2. Stakeholder Identification

Stakeholder	Interest / responsibility	Engagement
Plant manager / operations leadership	Production continuity, downtime, utilization, cost and productivity outcomes.	Review KPIs and release priorities.
Maintenance manager	Maintenance policies, schedules, work orders, team capacity and escalation rules.	Validate workflows and acceptance criteria.
Maintenance technicians	Receive alerts, inspect assets, perform and record maintenance.	Participate in story workshops and usability reviews.
Production supervisors	Line status, production impact, safe maintenance windows.	Validate downtime and scheduling constraints.
Reliability / condition-monitoring engineers	Health indicators, thresholds, failure modes and model validation.	Define indicators and verify risk outputs.
IT / OT engineers	Machine connectivity, data quality, network integration and system reliability.	Review integration and architecture.
Data engineers / data scientists	Telemetry pipelines, historical data, features and prediction validation.	Define data contracts and model evaluation.
Security / compliance team	Identity, permissions, audit logs, retention and industrial security requirements.	Review security controls and risks.
Product Owner	Represents business value, maintains backlog and accepts completed stories.	Own prioritization and sprint review decisions.
Scrum Master / facilitator	Facilitates Agile events and removes impediments.	Support planning, daily Scrum and retrospectives.
Developers and QA testers	Design, implement, test, document and integrate increments.	Estimate work and meet the Definition of Done.

3. Epics and User Stories

Stories use the required format: “As a <user>, I want <feature>, so that <benefit>.” The stories below are grouped by business capability. Acceptance criteria are provided for the selected first-sprint stories and key high-priority stories.

Epic ID / Epic	User story (ID)	Business value
E1 — Asset and Plant Management	US-01: As a maintenance manager, I want to register assets with plant, production line, asset ID, type and criticality, so that equipment is consistently identified.	Foundation for all asset workflows.
E1 — Asset and Plant Management	US-02: As a plant manager, I want to view assets by plant and production line, so that I can understand asset distribution.	Multi-site visibility.
E2 — Equipment Health Monitoring	US-03: As a reliability engineer, I want to view timestamped equipment health readings, so that I can identify abnormal operating conditions.	Visibility of equipment condition.
E2 — Equipment Health Monitoring	US-04: As an operations user, I want to see missing or stale readings flagged, so that I do not rely on outdated data.	Data trust and reliability.
E3 — Maintenance Management	US-05: As a technician, I want to record maintenance work, parts, time, status and findings, so that service history is traceable.	Reliable maintenance records.
E3 — Maintenance Management	US-06: As a maintenance manager, I want to assign work orders to technicians, so that responsibilities are clear.	Accountability and workload management.
E4 — Preventive Scheduling	US-07: As a maintenance planner, I want to configure maintenance intervals and due dates, so that preventive tasks are planned consistently.	Fewer overdue tasks.
E4 — Preventive Scheduling	US-08: As a production supervisor, I want to view proposed maintenance windows, so that servicing can be coordinated with production.	Reduced disruption.
E5 — Failure Risk and Alerts	US-09: As a reliability engineer, I want assets exceeding configured risk thresholds to be flagged, so that potential failures can be investigated early.	Earlier risk response.
E5 — Failure Risk and Alerts	US-10: As a technician, I want to receive an alert with asset, severity, timestamp and recommended next action, so that I can respond quickly.	Faster intervention.
E6 — Analytics and Reporting	US-11: As a plant manager, I want dashboards for downtime, maintenance status and asset utilization, so that I can monitor operations.	Operational visibility.
E6 — Analytics and Reporting	US-12: As an operations analyst, I want to export date-filtered maintenance reports, so that I can share evidence-based findings.	Reporting and decisions.
E7 — Security and Multi-Plant Administration	US-13: As an administrator, I want role-based access scoped by plant, so that users only access authorized information.	Secure multi-plant operation.
E7 — Security and Multi-Plant Administration	US-14: As an administrator, I want auditable changes to maintenance policies and critical records, so that important actions are traceable.	Governance and accountability.

4. Acceptance Criteria

The Given–When–Then criteria below are testable conditions for the first-sprint stories and selected follow-on stories. “Pass” means every criterion for the story is met.

US-01 — Register an asset

- AC-1: Given an authenticated user with asset-management permission, when they submit a unique asset ID and all required fields, then the system saves the asset and displays its assigned plant and line.
- AC-2: Given an asset ID already exists, when a user attempts to save another asset with that ID, then the system rejects the duplicate and explains the validation error.
- AC-3: Given a required field is missing or invalid, when the user submits the form, then no asset is created and field-level guidance is shown.

US-02 — View assets by plant and line

- AC-1: Given assets are assigned to different plants and lines, when an authorized user selects a plant or line filter, then only matching assets are displayed.
- AC-2: Given a user lacks access to a plant, when they request that plant's assets, then the system denies access and does not disclose the records.

US-03 — View health readings

- AC-1: Given readings exist for an asset, when an authorized user opens its health view, then readings are shown with metric name, value, unit and timestamp in chronological order.
- AC-2: Given no readings are available, when the health view is opened, then the system shows an explicit empty state rather than fabricated values.

US-04 — Flag stale readings

- AC-1: Given a configured freshness limit, when an asset's latest reading is older than that limit, then the asset is marked stale and the last-reading timestamp is visible.
- AC-2: Given a new valid reading arrives within the freshness limit, when the view refreshes, then the stale indicator is cleared.

US-05 — Record maintenance work

- AC-1: Given a valid asset and an authorized technician, when a maintenance record is submitted with required date, type, status and notes, then the record is saved and appears in the asset's history.
- AC-2: Given required fields are missing, when the record is submitted, then it is not saved and the user sees validation feedback.

US-07 — Configure preventive schedules

- AC-1: Given an authorized planner and a valid maintenance policy, when the planner saves an interval and next-due date, then the system stores the policy and displays the calculated due date.
- AC-2: Given the interval or date is invalid, when the planner saves, then the system rejects the policy and identifies the invalid field.

US-09 — Flag potential failure risk

- AC-1: Given a valid health indicator and a configured threshold, when the indicator crosses the threshold, then the asset is flagged with the rule, observed value and timestamp.
- AC-2: Given no validated predictive model is deployed, when risk is displayed, then the interface labels it as a rule-based indicator rather than a proven failure prediction.

US-10 — Notify maintenance team

- AC-1: Given a qualifying alert and configured recipients, when the alert is generated, then the system creates a notification containing asset ID, severity, time and recommended action.
- AC-2: Given delivery fails, when the notification service retries or reports the error, then the failure is logged and visible to an authorized administrator.

US-11 — View operations dashboard

- AC-1: Given a user has permission to a plant, when they open the dashboard, then KPI cards and charts reflect the selected plant and date range.
- AC-2: Given no data exists for the selected period, when the dashboard loads, then it shows a clear no-data state and does not imply zero failures unless the data supports that conclusion.

US-13 — Role-based plant access

- AC-1: Given a user is assigned a role and plant scope, when they access data or perform an action, then the system enforces the assigned permissions on the server.
- AC-2: Given an unauthorized user attempts a restricted action, when the request is processed, then access is denied and the event is logged where appropriate.

5. Product Backlog Prioritization

Prioritization uses a practical MoSCoW approach supported by relative business value (1–10), dependency order and risk reduction. Scores are illustrative planning inputs, not measured business facts. The Product Owner should confirm them with stakeholders.

Rank	Story	Priority	Value (1–10)	Dependency / rationale
1	US-01 Register assets	Must	10	Asset identity is prerequisite for health, maintenance and reporting.
2	US-03 View health readings	Must	10	Core monitoring capability; depends on asset records and a reading source.
3	US-05 Record maintenance work	Must	9	Creates traceable maintenance history.
4	US-07 Configure preventive schedules	Must	9	Enables planned maintenance and depends on asset records.
5	US-04 Flag stale readings	Must	8	Improves trust in monitoring data; depends on readings.
6	US-09 Flag potential risk	Must / next increment	8	High value, but depends on reliable readings and approved thresholds.
7	US-10 Send notifications	Should	8	Depends on alert generation and recipient configuration.
8	US-02 Filter assets by plant/line	Must	8	Supports multi-plant operations and role scope.
9	US-06 Assign work orders	Should	7	Builds on maintenance record and user/technician setup.
10	US-11 Operations dashboard	Should	7	Depends on trustworthy asset, reading and maintenance data.
11	US-13 Role-based plant access	Must	9	Cross-cutting security; foundational controls should be implemented early.
12	US-12 Export reports	Could	6	Useful after KPI definitions and source data are stable.
13	US-08 Coordinate maintenance windows	Should	6	Requires scheduling and production calendar information.
14	US-14 Audit critical changes	Should	7	Important for governance; scope depends on security design.

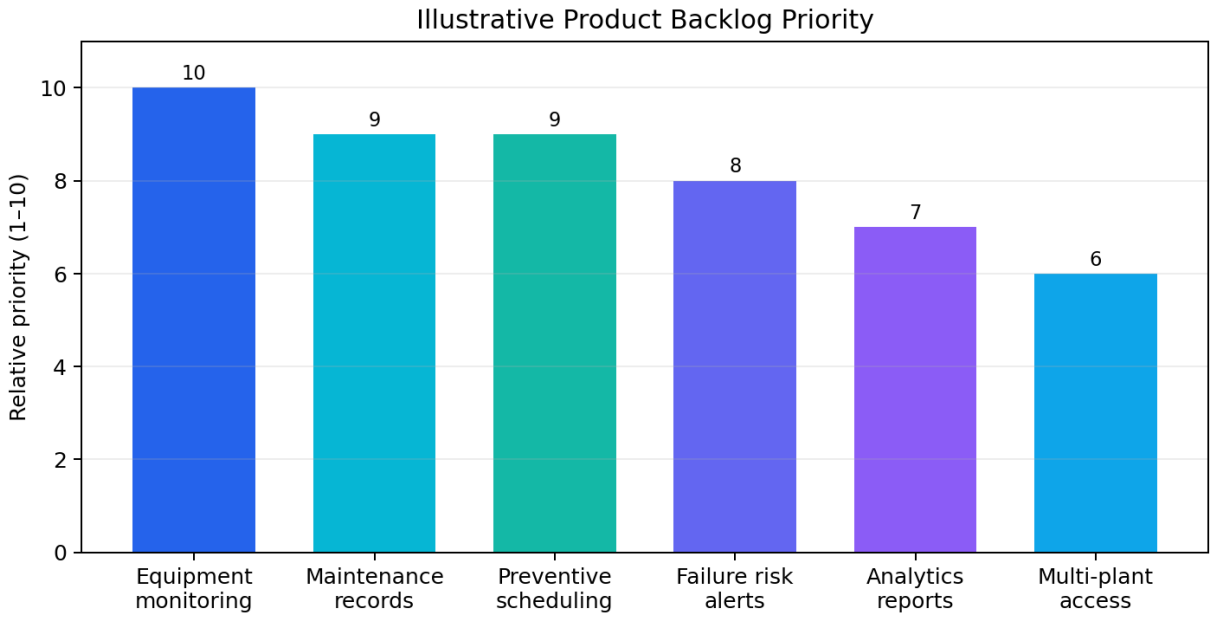


Figure 1. Illustrative relative priority scores for representative capabilities.

6. Sprint Planning

6.1 Sprint Assumptions

Planning item	Decision
Sprint duration	2 weeks (10 working days), an illustrative assumption.
Team	Product Owner, Scrum Master/facilitator, 2–4 developers, QA support; actual capacity must be confirmed.
Sprint 1 objective	Deliver a tested vertical slice for asset registration and viewing equipment health readings, with basic validation and documented data assumptions.
Capacity / velocity	No historical velocity is supplied. A conservative initial commitment is 16 story points, subject to team capacity and dependencies.
Definition of Done	Code reviewed; acceptance criteria passed; automated tests added where feasible; security checks performed; integrated in a test environment; documentation updated; no known critical defects.

6.2 Sprint Goal

Sprint Goal: Enable an authorized maintenance user to register an industrial asset and inspect its available, timestamped health readings, creating the foundation for later preventive scheduling and failure-risk alerts.

6.3 Selected Sprint Backlog

Story	Sprint scope	Points	Reason for selection
US-01	Register an asset with required fields, validation and duplicate protection.	3	Small, foundational, and testable.
US-03	Display available health readings with metric, value, unit and timestamp.	5	Delivers the first monitoring capability; integration/data assumptions add complexity.
US-05	Record maintenance activity with required fields and asset history.	3	Adds traceability, but can be deferred if health-data integration proves riskier.
US-07	Configure basic preventive schedule interval and next-due date.	5	Delivers an initial maintenance planning workflow.
Total	4 stories; 16 story points (provisional commitment).	16	Confirm after team discussion and capacity check.

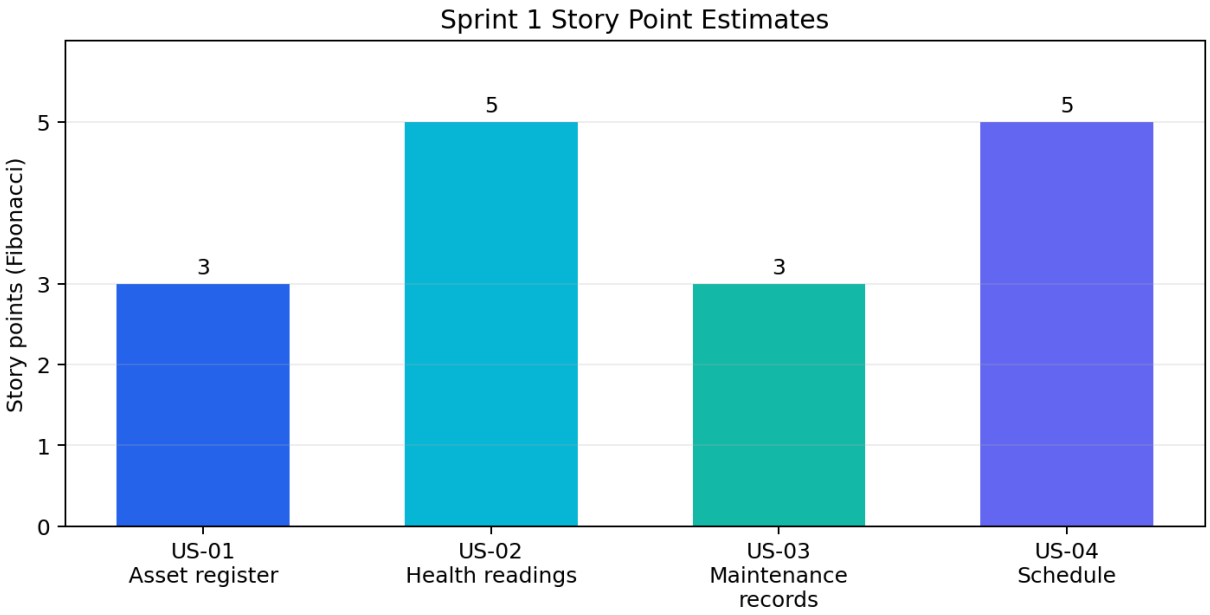


Figure 2. Proposed story point estimates for selected Sprint 1 stories.

6.4 Actionable Development Tasks

Story	Tasks	Owner role (suggested)
US-01	Confirm asset fields and unique key; design asset form/API; implement persistence and validation; prevent duplicate IDs; add unit/integration tests; demo registration flow.	Product Owner + Developer + QA
US-03	Define health-reading schema and units; identify/mock data source; implement reading retrieval; display metric/value/unit/timestamp; handle empty and stale data states; test ordering and errors.	Reliability Engineer + Developer + QA
US-05	Define maintenance record fields and statuses; implement create/list operations; link records to assets; validate required values; test history ordering and permissions.	Maintenance representative + Developer + QA
US-07	Agree basic policy fields; implement interval and due-date validation; calculate/display next due date; test valid/invalid policies and boundary cases.	Maintenance planner + Developer + QA
Cross-cutting	Set up repository, environment configuration and test database; implement authentication/authorization baseline; configure logging; conduct sprint review and retrospective; update user documentation.	Team

7. Story Point Estimation and Justification

Use Planning Poker with the modified Fibonacci sequence (1, 2, 3, 5, 8, 13). Each team member estimates independently, then discusses differences and agrees a team estimate. Story points represent relative effort, complexity, uncertainty and risk—not hours.

Story	Points	Justification	Main uncertainty
US-01	3	Limited form and persistence work with several validation rules; moderate testing.	Asset field agreement and unique-ID rules.
US-03	5	Requires a data contract, retrieval/display flow, timestamp/unit handling and empty/error states.	Availability and quality of telemetry data.
US-05	3	A straightforward create/list workflow with required-field checks and asset linkage.	Maintenance status definitions and permissions.
US-07	5	Policy validation and due-date logic need boundary testing and agreement on interval semantics.	Calendar rules, time zones and policy types.

The total is 16 story points. Because the team has no historical velocity in the problem statement, this is a provisional forecast rather than a guaranteed output. Re-estimate after technical discovery and use completed-sprint velocity to improve future planning.

8. Sprint Deliverables and Execution

8.1 Expected Deliverables

- An asset registration workflow with required-field validation and duplicate-ID protection.
- A health-reading view showing available readings with metric, value, unit and timestamp; empty/error states handled.
- A basic maintenance activity record linked to an asset, with history view.
- A basic preventive maintenance policy and next-due date workflow.
- Automated or documented test evidence for acceptance criteria.
- A working increment deployed to a test environment, plus setup and user notes.
- Updated product backlog with completed, incomplete and newly discovered items.

8.2 Two-Week Sprint Cadence

When	Activity	Output
Before Sprint Day 1	Backlog refinement; confirm acceptance criteria, dependencies, access and data assumptions.	Ready stories and clarified scope.
Day 1	Sprint Planning; agree goal, capacity and tasks.	Sprint backlog and shared commitment.
Daily	15-minute Daily Scrum focused on progress toward goal and impediments.	Adapted daily plan.
Days 2–8	Design, build, integrate and test; refine with Product Owner as needed.	Potentially releasable increment.
Days 9–10	Complete tests, fix defects, demo and document; Sprint Review and Retrospective.	Accepted increment, feedback and improvement actions.

8.3 Definition of Done

- Every acceptance criterion for a completed story passes.
- Code is reviewed and merged; tests run successfully in the agreed environment.
- Input validation, error handling, access control and logging are addressed for the delivered scope.

- No open critical or high-severity defect blocks the feature.
- Configuration and setup instructions are documented; no secrets are committed to source control.
- The Product Owner reviews the increment and confirms it meets the agreed acceptance criteria.

9. Risks, Dependencies and Mitigations

Risk / dependency	Impact	Mitigation
Telemetry is unavailable or inconsistent	Health-monitoring story may be blocked or display misleading data.	Agree schema early; use clearly labelled synthetic/sample data for the prototype; show source and timestamp.
Asset identifiers differ between plants	Duplicate records and incorrect maintenance links.	Define a global identifier strategy and plant/line relationships.
No baseline maintenance KPIs	Business improvements cannot be quantified.	Collect baseline downtime, maintenance cost, failure count and utilization before claiming benefits.
Risk thresholds are not validated	False alarms or missed failures.	Start with configurable rule-based flags; validate predictive models against historical labelled data before operational use.
Unauthorized cross-plant access	Confidentiality and operational security risk.	Enforce server-side role and plant scopes; test both allowed and denied access.
Team capacity is unknown	Sprint commitment may be unrealistic.	Confirm availability and dependencies in planning; reduce scope before sacrificing quality.

10. Expected Outcomes and Evaluation

The platform is expected to improve visibility and maintenance planning and, over time, help reduce unexpected failures and downtime. These are intended outcomes, not guaranteed results of Sprint 1. Evaluate impact against an agreed pre-deployment baseline.

Outcome	Suggested KPI	Evaluation approach
Reduced unexpected failures	Unplanned failure events per operating hour or month	Compare equivalent equipment groups and operating periods.
Lower downtime	Unplanned downtime hours per asset/line	Compare baseline with post-rollout, accounting for production volume.
Lower maintenance cost	Maintenance spend per asset or operating hour	Separate planned, corrective and emergency work.
Improved asset utilization	Available operating time / planned operating time, using an agreed formula	Report formula and exclusions consistently.
Better maintenance planning	On-time preventive maintenance completion rate	Track due tasks completed by due date.
Better alerts	Alert precision, recall, lead time and acknowledgement time	Validate against confirmed failure events and technician feedback.

11. Assessment Rubric Mapping

Rubric criterion	Weight	Evidence in this document	Excellent-level check
Project Vision & Stakeholder Identification	30%	Sections 1–2	Clear vision, aligned objectives, relevant stakeholders and roles.
Epics & User Stories	25%	Section 3	Seven epics and fourteen user stories using the required format.
Acceptance Criteria	20%	Section 4	Measurable Given–When–Then conditions for key stories.
Product Backlog Prioritization & Sprint Planning	15%	Sections 5–6	Ranked backlog, dependencies, selected sprint scope and actionable tasks.
Story Point Estimation, Sprint Goal & Presentation	10%	Sections 7–8; figures 1–2	Fibonacci estimates justified; clear goal, deliverables and professional layout.

12. Conclusion

This Agile plan turns the smart-manufacturing problem into a prioritized, testable delivery approach. The first sprint focuses on foundational asset and maintenance workflows while establishing a reliable path to equipment health monitoring and preventive scheduling. Later sprints can extend the increment with risk alerts, notifications, dashboards, reporting, stronger multi-plant administration, and validated predictive models. Regular stakeholder feedback, explicit acceptance criteria, data-quality controls and evidence-based KPI measurement will help keep the product aligned with manufacturing needs.

Appendix A — Suggested Future Sprint Sequence

Sprint / increment	Indicative focus	Outcome
Sprint 2	Plant/line filtering, stale-data indicators, role/plant authorization hardening.	More trustworthy monitoring and safe multi-plant access.
Sprint 3	Risk thresholds, alert lifecycle, notification delivery and acknowledgement.	Actionable early-warning workflow.
Sprint 4	Operations dashboards, date filters, maintenance reports and exports.	Centralized operational visibility.
Sprint 5+	Historical data pipeline, model evaluation, model monitoring, asset lifecycle and external system integrations.	Progressively validated predictive capabilities and broader scalability.

Planning note: The sequence is indicative and should be re-prioritized after each Sprint Review based on user feedback, technical learning, data readiness and business value.